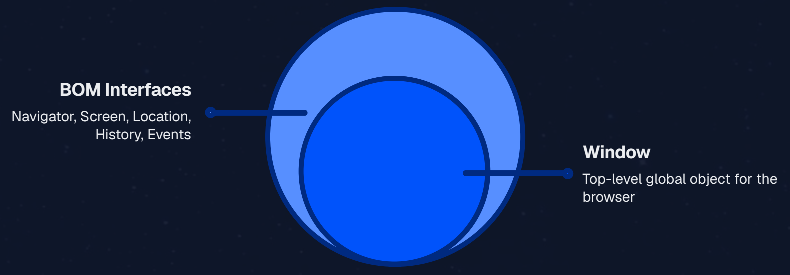
DEVELOPER CONFERENCE

Browser Object Model

BOM

Client Side Technologies — JavaScript



Created by
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What is DOM?

DOM

Document Object Model — a W3C standard for accessing and manipulating HTML documents

W3C Standard

Officially standardized by the World Wide Web Consortium

Platform Independent

Works across all operating systems and platforms

Language Independent

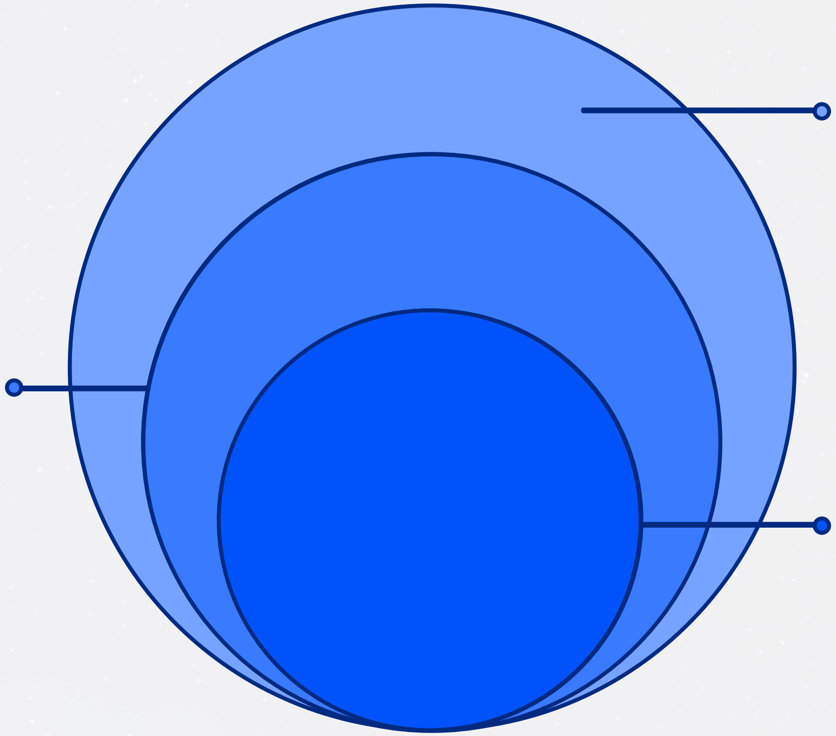
Can be used with JavaScript, Python, or any other language

Defines

A standard way to access and manipulate HTML documents

What is BOM?

Browser Object Model — covers objects relating to the browser



Each child object relates directly to the window object.

The DOM resides within the BOM.

DOM Object Hierarchy

Hierarchical tree diagram of the browser's object structure.

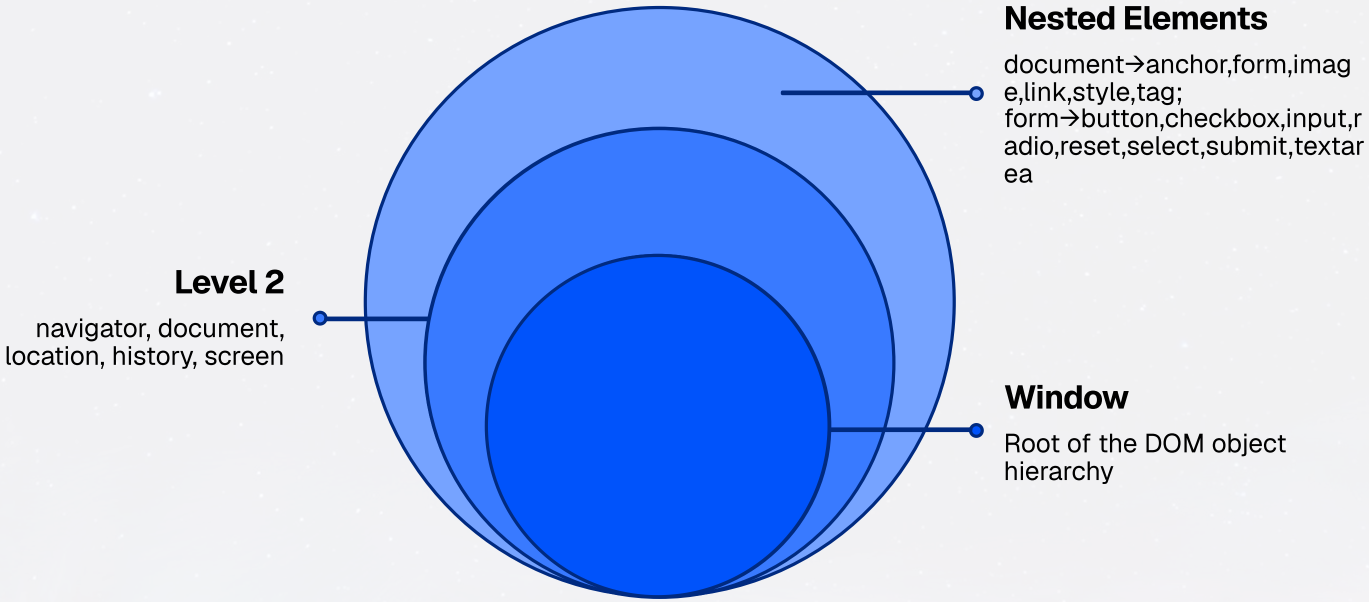
window

root

navigator • document • location • history • screen

document children: anchor • form • image • link • style • tag

form children: button • checkbox • input • radio • reset • select • submit • textarea



Window Object — Properties

Property	Description	Syntax
innerHeight	Returns the inner height of a window's content area	window.innerHeight
innerWidth	Returns the inner width of a window's content area	window.innerWidth
outerHeight	Returns the outer height of a window, including toolbars/scrollbars	window.outerHeight
outerWidth	Returns the outer width of a window, including toolbars/scrollbars	window.outerWidth
screenLeft / screenX	Returns horizontal coordinate of the window relative to the screen	window.screenLeft
screenTop / screenY	Returns vertical coordinate of the window relative to the screen	window.screenY
pageXOffset / scrollX	Pixels the document has been scrolled horizontally	window.pageXOffset
pageYOffset / scrollY	Pixels the document has been scrolled vertically	window.pageYOffset

Window Object — Methods

Method	Description	Syntax
alert()	Displays an alert box with a message and OK button	window.alert('Hello')
confirm()	Dialog with OK (true) and Cancel (false)	window.confirm('Exit?')
prompt()	Dialog prompting the user for input	name = prompt('Name','')
open()	Opens a new browser window	window.open(url, name, props)
close()	Closes a specified window	window.close()
focus()	Sets focus to the current window	window.focus()
blur()	Removes focus from the current window	window.blur()
print()	Prints the contents of the window	window.print()
stop()	Stops the window from loading	window.stop()
setInterval()	Evaluates an expression at specified intervals	setInterval(fn, 500)
clearInterval()	Clears a time interval set with setInterval	clearInterval(t)

Screen Object

The screen object provides information about the desktop outside the browser

[window.]screen

🖥️ Screen dimensions and display properties

Property	Description
availHeight	Returns the height of the screen (excluding Windows Taskbar)
availWidth	Returns the width of the screen (excluding Windows Taskbar)
colorDepth	Returns the bit depth of the color palette for displaying images
height	Returns the total height of the screen
pixelDepth	Returns the color resolution (in bits per pixel) of the screen
width	Returns the total width of the screen

Navigator Object

Represents the browser application. All properties are read-only.

[window.]navigator

Property	Description	Syntax
appName	Get the name of the browser	navigator.appName
appVersion	Get the version of the browser	navigator.appVersion
language	Get the language of the browser	navigator.language
cookieEnabled	Returns whether the browser allows cookies	navigator.cookieEnabled
platform	Return the name of the OS	navigator.platform
onLine	Determines whether the browser is online	navigator.onLine
geolocation	Returns a Geolocation object to locate the user's position	navigator.geolocation

Method: `javaEnabled()` — returns true if Java is enabled in the browser

Location Object

The Location object is part of the Window object and refers to the current URL

Properties

href

Sets or returns the entire URL (default property)

hash

Sets or returns the anchor part (#) of a URL

search

Sets or returns the querystring part of a URL

Methods

replace(URL)

Replaces current document; doesn't add to history

assign(URL)

Loads new URL and adds an entry to browser history

reload()

Reloads the current document

Code example

```
// Get current URL
var url = location.href;

// Replace without history entry
location.replace('https://example.com');

// Reload the page
location.reload();
```

History Object

The history object is an array of URLs the client has previously visited

[window.]history

Property

length

Returns the number of URLs in the history list

Methods

back()

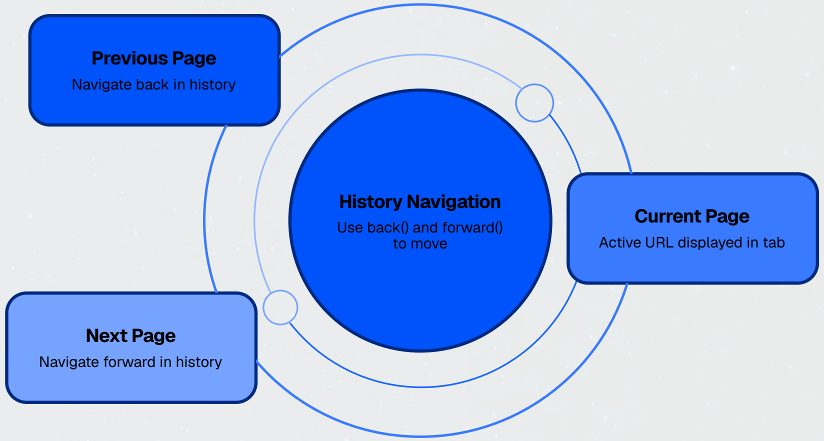
Loads the previous URL in the history list (equivalent to clicking Back)

forward()

Loads the next URL in the history list (equivalent to clicking Forward)

go(index)

Loads a specific URL from the history list; go(-1) = back, go(1) = forward



Use **Back** and **Forward** to move through previously visited pages, or **go(index)** to jump to a specific point in the history stack.

Event Object

DOM events allow JavaScript to register event handlers on HTML elements. When an event fires, an Event object is created holding extra information about that event.

Event Object Properties

Property	Description
target	The element that fired the event
type	Type of event (e.g., 'click', 'mouseover')
timeStamp	Time in milliseconds since Jan 1 1970 at which the event was created
bubbles	Returns whether the event is a bubbling event
cancelable	Returns whether the event's default action can be prevented

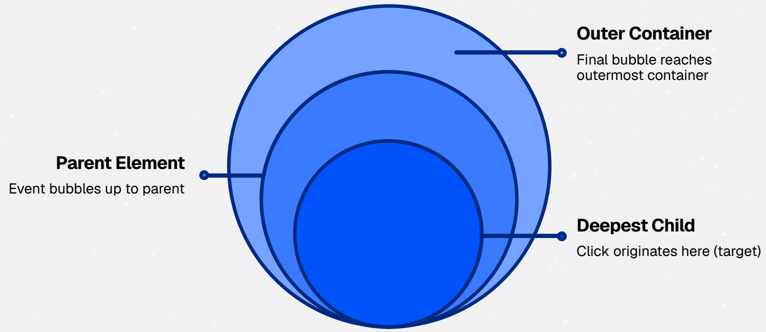
Key Methods

- **preventDefault()** — cancels the default action
- **stopPropagation()** — stops event from bubbling up the DOM

Event Propagation (Bubbling) & Binding

Event Bubbling

After an event triggers on the deepest element, it triggers on parents in nesting order



Use stopPropagation() to cancel bubbling (W3C browsers)

Use e.cancelBubble = true for IE < 9

```
function handler(e) {
  e.stopPropagation(); // W3C
  e.cancelBubble = true; // IE<9
}
```

Binding Events

1. Tag Attribute

```
onclick="myFunc()"
```

2. Object Property

```
element.onclick = myFunc; // no ()
```

3. addEventListener()

```
el.addEventListener("click", fn)
```

addEventListener() advantages:

- Doesn't overwrite existing handlers
- Easier bubbling / capturing control
- Separates JS from HTML markup
- Remove with removeEventListener()

MouseEvent & KeyboardEvent Properties

MouseEvent Properties

screenX / screenY

Coords relative to the screen

clientX / clientY

Coords relative to the browser window

pageX / pageY

Coords relative to the document

offsetX / offsetY

Coords relative to the target element

button

0=Left, 1=Middle, 2=Right mouse button

altKey / ctrlKey / shiftKey

True if modifier key was also pressed

detail

How many times mouse was clicked

movementX / movementY

Coords relative to last mousemove event

KeyboardEvent Properties

key

String representing the key value (preferred)

code

String with the physical key code

which (deprecated)

Numeric key code — use key instead

altKey / ctrlKey / shiftKey

True if modifier key was pressed

Latest DOM Events Spec recommends using the key property instead of which or keyCode

Thank You

Browser Object Model & DOM Mastery

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